

May 10, 2026



Dear Awards Committee,

I am pleased to write in strong support of Martin Bissonnette's application for the Co-Op Student of the Year Award.

Martin joined Irradiant Technologies to support the work of a \$14 million DARPA-funded contract focused on the development of new approaches to manufacturing 3D non-planar microsystems. His primary role was on the materials team, where he was responsible for designing and printing conductive silver structures.

Soon after starting his work term, Martin quickly became one of the primary operators of our point-scan lithography system, a platform built of custom hardware and software that Martin actively contributed to maintaining and troubleshooting. During this time, he independently implemented a stitching feature that enabled the printing of large structures by combining multiple fields of view, a capability that did not exist prior to his arrival. He also developed a visualizer and produced complete documentation for file generation to support the stitching workflow. Martin continued to perform prints for our DARPA milestones while developing these tools, and conducted surface roughness measurements that directly allowed for the securement of a \$1.8 million milestone.

Alongside another co-op student, Martin also implemented the automation of multiple post-processing steps using a pipetting robot, reducing our experimental process time from one week to only three days. Specific contributions included designing and 3D printing hardware fixtures used by the robots, and evaluating the chemical compatibility of these fixtures with our processing reagents. The resulting automation set-up has since been extended beyond the DARPA project, and is used company wide to increase turnover time of other tasks.

In addition to his core responsibilities, Martin was actively involved in several other departments. He was responsible for designing and printing of RF antennas to support a new government grant application. Martin's designs continue to be used as demonstrators for several grant applications. He was also responsible for the optical design implementation for a large customer-facing project, working closely with both metrology and optics teams to achieve this.

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Perhaps one of Martin's most long-lasting contributions came in an area entirely separate from his main duties. Fine processing and precision cutting of our nanoporous glass scaffold has been a long standing challenge for the company. The material is brittle and fragile by nature, and precise cutting or dicing often results in low yields and long process times. Martin independently researched this problem and proposed the use of a low-cost UV dicing tool capable of making such precision cuts within the scaffold. The process has since been extended to the whole of the company, and has become our standard operating procedure.

In his eight months, Martin consistently performed far above the level expected of a co-op student. He is an excellent communicator, a reliable and supportive teammate, and has a skill for finding and implementing practical solutions. Overall, he has been a pleasure to work with, and I recommend him for this award without reservation.

Sincerely,

Sheilan Sinjari
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